BMEN 600 | TEAM PROJECT | FINAL PROJECT

FINAL PROJECT

Final Research Paper: 30% of the course grade

Final Oral Presentation: 10% of the course grade

Team: 3-4 students

Due: Submit to D2L by Friday, December 4 at 11:59 p.m.

Length: Maximum 10 pages, excluding title page and references

Submit: One PDF including one link to your GitHub repository

Project Overview

The final project is designed as an AI-assisted biomedical research experience. Teams are **encouraged** to use AI tools to support literature searches, coding, data analysis, troubleshooting, and communication. Our goal is to use powerful AI tools to help us conduct a meaningful analysis of a biomedical engineering dataset. Your team will remain responsible for verifying sources, testing code, checking outputs, justifying research decisions, and ensuring that the conclusions are supported by the data.

For this course, an **original analysis** means that your team develops and investigates its own focused research question and produces its own results and interpretation. You are not required to invent a new algorithm or establish publication-level novelty. A meaningful contribution might include:

- investigating a new or underexplored question using an existing dataset;
- comparing groups, measurements, or analytical approaches;
- examining a relevant subgroup, limitation, or source of error;
- reproducing a previous analysis with an important improvement;
- evaluating whether a published finding holds under different conditions; or
- reporting an informative negative or inconclusive result.

Final Project Deliverables

The final paper will extend your work completed for your midterm plan. You should use the feedback received on your midterm plan to revise and improve your work. Reasonable changes from your midterm plan are allowed, if justified. You are allowed to copy and paste work from your midterm plan to support your research paper, although the research paper should describe your completed work and interpretation.

1. **Research Paper (PDF).** Write the report as a cohesive research paper. The descriptions below identify the required content, but they should not be copied into the paper or answered individually as a list of questions.
2. **Final GitHub Repository (GitHub link).** Share a link to your organized project materials.
3. **Final Oral Presentation.** A 5-minute group presentation sharing the outcome of your project.

Research Paper

Section 1 – Abstract (~ 1/2 page)

In approximately 200–250 words summarize: the biomedical problem and research question; the dataset and study population or samples; the main analytical approach; the most important result; the interpretation of that result and main findings, and the main limitations.

Section 2 - Introduction (1.5 - 2 pages)

Introduce the biomedical challenge and explain why it is important. Identify the population or end-users affected by the problem. Synthesize the most relevant background research. Explain what is currently known, how the problem has previously been studied, and the important findings, methods, limitations, or disagreements in the literature. Give particular attention to the studies most closely related to your research question. Conclude the Introduction by identifying what remains unresolved, stating the focused research question, and explaining the specific contribution the project is intended to make.

Section 3 - Dataset and Methods (~ 2 pages)

Describe the completed study in enough detail that another student with appropriate access to the data could understand and reproduce the analysis.

Dataset:

Identify the dataset name, source, date accessed, version if available, and any publication that introduced or validated it. Describe:

- the population, participants, specimens, images, signals, or other samples represented;
- what constitutes one independent case;
- the important variables, measurements, groups, labels, or outcomes;
- whether one person or specimen contributes multiple records;
- the final inclusion or exclusion decisions; and
- the number of cases or observations included in the analysis.

Discuss responsible use of the dataset, including applicable data-use terms, privacy or re-identification concerns, restrictions on sharing, and important bias or representation limitations.

Computational Workflow:

Describe the major steps that your team took in a logical order, such as:

- accessing and loading the data;
- checking data quality;
- cleaning, excluding, or preprocessing records;
- calculating measurements or extracting features;
- conducting the main statistical, computational, or modelling analysis;
- making a meaningful comparison; and
- calculating uncertainty, variability, or evaluation measures.

Identify the software, programming language, packages, toolboxes, platforms, or computational tools used. Explain the purpose of the main scripts and how they work to produce your results. If an AI-based tool directly labeled, transformed, analyzed, or modelled the data, describe its role as part of the Methods.

Evaluation:

Identify the primary result used to answer the research question and explain why it is appropriate for the biomedical problem.

Describe the comparison used to place the result in context. Depending on the project, this might be:

- a comparison between groups;
- a simple or established analytical approach;
- a relevant reference value;
- current practice;
- a baseline predictive method; or
- a method reported in a closely related study.

Section 4 - Results (~ 2 pages)

Present the evidence produced by the completed analysis. Organize this section around the research question and major findings rather than describing the order in which the team performed the work.

Present:

- the primary result that answers the research question;
- the meaningful comparison identified in your Methods;
- appropriate variability or uncertainty;
- any important secondary, subgroup, sensitivity, or error analysis that is feasible for the project; and
- relevant negative, inconclusive, or unexpected findings.

Include at least:

- one table or figure describing the dataset, study sample, or data quality; and
- one table or figure presenting the primary result and its comparison or uncertainty.

Every table and figure must be numbered, captioned, discussed in the text, and generated from the team's analysis. Do not hide an unexpected or negative result. A carefully analyzed result that does not support the original expectation can still be scientifically informative.

Section 5 – Discussion (~ 1.5 - 2 pages)

Begin by answering the research question using the evidence presented in the Results.

Interpret the findings in relation to the biomedical problem and the previous research discussed in the Introduction. Explain where the findings agree with, extend, complicate, or differ from the closest studies.

Discuss what the results could mean for the affected population, end-users, or biomedical field. Keep these implications proportionate to the dataset and study design.

Identify the most important limitations, and where relevant, plausible alternative explanations for these findings. These may involve sampling, representation, missing data, measurement quality, repeated records, study design, statistical uncertainty, model evaluation, computational constraints, or the limits of the dataset.

Conclude the Discussion by identifying the most useful next scientific or technical step.

Section 6 – Conclusion (~ 0.5 page)

Provide a concise, evidence-based conclusion that states: the answer to your research question; the project's main contribution; and the most important qualification or limitation. Do not introduce new results or make unsupported claims about causality, clinical usefulness, generalizability, or readiness for deployment.

Data and Code Availability

Provide the dataset source and GitHub repository link. Explain any restrictions that prevent the dataset or other materials from being shared.

AI-Assisted Work

Identify the main AI tools used and the substantive purposes for which they were used. Briefly explain how team members verified AI-assisted references, code, analyses, or writing. Full chat histories are not required.

Author Contributions

State each team member's main contributions to the research question, literature review, data preparation, analysis, code, figures, interpretation, writing, and revision.

References

Use IEEE citation style.

Final GitHub Repository

Your final GitHub repository should include:

- **README file.** State the research question, identify the dataset, explain how to access the data, provide a map of the files, and give the steps required to run the analysis.

- **Analysis code.** Include the scripts or notebooks used to clean or preprocess the data, perform the main analysis, and generate the reported tables or figures.
- **Software and setup instructions.** Identify the platform, programming language, and required packages or toolboxes. A requirements.txt, environment.yml, MATLAB requirements list, or clearly explained Colab setup is sufficient.
- **Organized files.** Use understandable folder and file names. Separate code, results, figures, and documentation where appropriate.
- **Expected outputs.** Explain which script or notebook generates the primary result and the main paper figures or tables.
- **Submitted version.** Provide the commit identifier or commit link for the version being graded.

Final Oral Presentation

Each team will give a **5-minute presentation** followed by approximately **5 minutes** of questions.

The presentation should include:

- the biomedical problem and research question;
- the dataset and independent case;
- the completed analysis workflow and meaningful comparison;
- the results;
- limitations and next steps;
- and the final interpretation and biomedical relevance

Every team member should speak for at least 1 minute. All team members must be able to explain the project's research question, data, methods, results, limitations, and their own contributions.

Research Paper Rubric

The final written report is marked out of 21 points. Each criterion is worth 3 points.

1. Introduction - 3 points

3 - Excellent. The Introduction clearly establishes a meaningful biomedical challenge, why it matters, and the population or end-users affected. It synthesizes relevant scholarly sources to explain what is currently known, how the problem has been investigated, and the important findings, methods, limitations, or disagreements. The closest prior studies are used to establish a specific unresolved problem. The research question and project contribution follow logically from this evidence.

2 - Good. The biomedical challenge and general research direction are credible, but at least one element such as the significance, affected population, research question, or potential contribution requires clarification.

1 - Developing. The Introduction is primarily driven by the availability of a dataset or organized mainly as a series of individual paper summaries. The relationship among the biomedical problem, previous research, unresolved problem, research question, and project contribution is weak.

0 - Not demonstrated. No coherent biomedical research context, evidence base, or focused research question is established.

2. Dataset and Methods - 3 points

3 - Excellent. The dataset is clearly described and identified. The samples, independent case, important variables, and outcomes are accurately described. The discussion meaningfully addresses data use terms, privacy concerns, bias, and representation limitations in relation to the project. The Methods clearly describe the completed analysis and follow logically from the research question. The major analytical steps, meaningful comparison, and approach to variability or uncertainty are appropriate and sufficiently described to understand and reproduce the analysis. The software and computational tools used are clearly identified.

2 - Good. The dataset appears usable, but at least one important detail needs clarification. The overall workflow is appropriate and understandable, but at least one important analytical, computational, comparison, uncertainty, or evaluation decision requires further explanation or justification. Some information needed to reproduce the methods may be incomplete.

1 - Developing. Important information about the dataset is missing or misunderstood. A substantial concern involving the independent case, repeated records, sample selection, variables, data permissions, privacy, bias, or representation may affect the interpretation or validity of the study. The workflow is vague, incomplete, only loosely connected to the research question, or contains a major methodological issue that could produce a misleading result. The role of the code, computational tools, comparison, or evaluation approach is difficult to determine.

0 - Not demonstrated. The dataset is not meaningfully described, cannot support the reported research question, or cannot be used legally or ethically as presented. No assessable or usable analysis and evaluation method is presented.

3. Results - 3 points

3 - Excellent. The Results are organized around the research question and major findings. The report presents the primary result, a meaningful comparison, and appropriate variability or uncertainty, along with relevant secondary, subgroup, sensitivity, error, negative, or inconclusive findings where feasible. At least one team-generated table or figure describes the dataset or data quality, and at least one presents the primary result. Tables and figures are accurate, readable, numbered, captioned, and discussed in the text. The reported results are consistent with the Methods.

2 - Good. The primary result and relevant team-generated outputs are present and generally interpretable. However, the comparison, uncertainty, completeness, organization, visual presentation, or connection to the research question requires further development.

1 - Developing. The Results are incomplete, difficult to interpret, organized mainly as a chronological record of the team's work, or weakly connected to the stated research question and Methods. Important comparisons, uncertainty, or supporting evidence are missing.

0 - Not demonstrated. No meaningful results are presented, or the reported results cannot be supported by the submitted analysis.

4. Discussion - 3 points

3 – Excellent. The Discussion provides a clear, evidence-based answer to the research question. It interprets the findings in relation to the biomedical problem and the closest prior studies and explains their potential relevance to the affected population, end-users, or biomedical field.

Important limitations and plausible alternative explanations are addressed. The discussion clearly distinguishes what the study demonstrates from what it only suggests or cannot establish and identifies an appropriate next step.

2 - Good. The interpretation is generally reasonable and connected to the results, but the relationship to previous research, biomedical meaning, limitations, boundaries of the claims, or next step requires further development.

1 – Developing. The Discussion mainly repeats the Results, provides limited interpretation, overlooks important limitations, or makes claims that extend substantially beyond the dataset and study design.

0 - Not demonstrated. No meaningful interpretation of the results or defensible answer to the research question is provided.

5. Abstract and Conclusion - 3 points

3 - Excellent. The abstract concisely and accurately summarizes the biomedical problem, research question, dataset, analytical approach, primary result, interpretation, and principal limitation. The conclusion directly answers the research question, states the project's evidence-supported contribution, and identifies an appropriate qualification or limitation. Both sections are consistent with the evidence presented in the paper and avoid introducing new results or overstating the findings.

2 - Good. The abstract and conclusion communicate the overall study accurately, but one important element is missing, unclear, disproportionate, or insufficiently connected to the reported evidence.

1 - Developing. The abstract or conclusion is substantially incomplete, overly general, inconsistent with the paper, or makes claims that are not adequately supported by the results.

0 - Not demonstrated. The abstract and conclusion are missing or do not provide an assessable summary and conclusion for the completed study.

6. AI Research and GitHub Repository - 3 points

3 - Excellent. The Data and Code Availability, AI-Assisted Work, and Author Contributions statements are complete and specific. The GitHub link works and the repository contains a clear README, dataset-access instructions, software and setup information, organized analysis code, and an explanation of which scripts or notebooks generate the main reported outputs. Another authorized person could identify how the analysis was conducted and reproduce a central table or figure without undocumented instructions.

2 - Good. The required statements and repository contain most of the expected information and materials. However, some data access, setup, organization, output-mapping, AI-verification, contribution, or reproducibility information remains unclear or incomplete.

1 - Developing. The repository is present but incomplete, difficult to understand, or not readily reproducible. Required transparency statements are vague or incomplete, or individual contributions cannot be clearly identified.

0 - Not demonstrated. The repository is missing or inaccessible, no assessable analysis code is provided, or the required availability, AI-use, and contribution information is absent.

7. Scholarly Writing and Organization - 3 points

3 - Excellent. The report is written as a cohesive research paper. Ideas are organized logically across sections. Tables and figures are clear, numbered, captioned, and discussed in the text. Claims are supported with appropriate citations, and IEEE referencing is consistent. Writing errors are minimal and do not interfere with clarity.

2 - Good. The report is understandable and generally organized, but some sections are repetitive, weakly connected, or written as separate answers to assignment prompts. Some tables, figures, claims, or references are not fully integrated or consistently formatted. Writing or formatting issues occasionally affect clarity.

1 - Developing. The final report is difficult to follow because of recurring problems with organization, paragraph structure, transitions, or clarity. Important claims are unsupported, or tables, figures, and references are poorly integrated.

0 - Not demonstrated. The submission does not communicate the project in a usable scholarly format. Substantial problems with writing, organization, citations, or presentation prevent the reader from understanding or evaluating the work.

Oral Presentation Rubric

The oral presentation will be marked out of 15 points. Each criterion is worth 3 points.

1. Biomedical Research Question - 3 points

3 – Excellent. The presentation clearly establishes the biomedical problem, why it matters, and the population or end-users affected. The research question is focused and clearly connected to the purpose and contribution of the completed project.

2 – Good. The biomedical problem and research question are generally understandable, but the significance, affected population, project purpose, or connection between the problem and question requires some clarification.

1 – Developing. The project is introduced broadly or mainly in terms of the available dataset or analytical technique. The biomedical significance or research question is unclear or difficult to identify.

0 – Not demonstrated. No coherent biomedical problem or research question is presented.

2. Data Analysis and Workflow - 3 points

3 – Excellent. The presentation clearly identifies the dataset, population or samples, independent case, and important variables or outcomes. The major analysis steps and meaningful comparison are explained accurately and at an appropriate level of detail. The audience can understand how the analysis was used to answer the research question.

2 – Good. The dataset and general analysis workflow are understandable, but one important element—such as the independent case, variables, preprocessing, analytical steps, comparison, or evaluation approach—requires clarification.

1 – Developing. The dataset or workflow is described vaguely, inaccurately, or in excessive or insufficient detail. The connection between the data, analysis, and research question is difficult to follow.

0 – Not demonstrated. No assessable description of the dataset or analysis workflow is provided.

3. Results and Discussion - 3 points

3 – Excellent. The team clearly presents the primary result and provides an evidence-based answer to the research question. The biomedical meaning and relevance of the findings are clearly explained. The interpretation is proportionate to the dataset and study design. Important limitations are acknowledged, the boundaries of the conclusions are clear, and the proposed next step follows logically from the completed work.

2 – Good. The interpretation is generally reasonable, but the biomedical relevance, limitations, boundaries of the claims, or next step requires further development.

1 – Developing. The interpretation mainly repeats the results, overlooks important limitations, or makes claims that extend beyond the evidence. The proposed next step is vague or disconnected from the study.

0 – Not demonstrated. No meaningful interpretation, limitation, or next step is provided.

4. Organization and Communication - 3 points

3 – Excellent. The presentation follows a clear and coherent scientific storyline. Information is selected appropriately for the 5-minute time limit, transitions are effective, and terminology is suitable for the intended scientific audience. Slides are readable, visually consistent, and focused on evidence. Delivery is clear, appropriately paced, and completed within the allotted time.

2 – Good. The presentation is generally organized and understandable, but some content, transitions, slides, terminology, pacing, or timing occasionally reduces clarity.

1 – Developing. Recurring problems with organization, slide readability, amount of detail, pacing, transitions, or timing make the presentation difficult to follow. Important content may be omitted because time is not managed effectively.

0 – Not demonstrated. The presentation is not delivered in an assessable or understandable form.

5. Participation and Questions - 3 points

3 – Excellent. Every team member has a substantive speaking role of at least 1 minute. Team members demonstrate shared understanding of the research question, dataset, methods, results, limitations, and individual contributions. Responses to questions are direct, accurate, evidence-based, and appropriately cautious when uncertainty remains.

2 – Good. Every team member participates, but speaking time or project understanding is somewhat uneven. Responses to questions are generally accurate but contain minor gaps in explanation.

1 – Developing. Participation is substantially uneven, and team members have difficulty explaining important parts of the project or their own contributions. Responses to questions are vague, incomplete, or require additional prompting.

0 – Not demonstrated. The team does not demonstrate shared participation, or responses to questions are inaccurate.